MILITARY SPECIFICATION

CUSHION, FORWARD M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for aft cushions used in M753, 8 Inch Projectiles (see 6.1).

1.2 Classification. Forward cushions covered by this specification shall be of the following types (see 6.2):

Type I - Part No. 9299055-1 (Fabricated from Soft Grade Silicone Sponge Rubber)

Type II - Part No. 9299055-2 (Fabricated from Medium Grade Silicone Sponge Rubber

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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SPECIFICATIONS

MILITARY

MIL-R-6130	Rubber, Cellular, Chemically Blown
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specifications for

STANDARDS

MILITARY

MIL-STD-286	Propellant, Solid, Sampling Examinations and Tests
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9299055	Cushion, Forward
9328352	Sponge Rubber, Silicone (SOFT GRADE)
9328353	Sponge Rubber, Silicone (MEDIUM GRADE)

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 792	Specific Gravity and Density of Plastics by Displacement
ASTM D 2240	Rubber Property, Durometer Hardness Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 191030.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The forward cushions shall comply with all applicable requirements specified on drawing (dwg.) 9299055, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. Silicone sponge rubber used in fabricating forward cushions shall comply with all requirements specified on dwg. 9328352 or dwg. 9328353, as applicable, and with all requirements specified herein. The contractor shall submit test samples of the materials to the procuring agency for analysis and approval in accordance with the requirements of this specification prior to use in fabrication of forward cushions (see 6.3).

3.3 Material properties.

3.3.1 Physical properties. The physical properties of the silicone sponge rubber at 73 ± 3°F (23 ± 3°C) shall be as specified in Table I. Testing shall be performed in accordance with 4.5.1 thru 4.5.3, as applicable.

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TABLE I. Physical properties.

Properties	Soft Grade	Medium Grade
Density at 73°F, max. (lb/cu in)	.013	.020
Hardness, Shore A	3-5	6-8
Compression deflection, psi (compressed at 75% of original thickness at room temp.)	2-7	6-14
Compression set, % max (compressed to 50% of original thickness for 22 hours at 212°F)	15	15

3.3.2 Chemical composition. The silicone rubber shall contain a minimum of 60% organic material by weight for soft grade material, and a minimum of 55% organic material by weight for medium grade material. Testing shall be performed in accordance with 4.5.4.

3.3.3 Infrared spectrum. The infrared spectrum of the silicone sponge rubber shall qualitatively match the applicable wavelength characteristics of Figure 1 or 2, as applicable. Testing shall be performed in accordance with 4.5.5.

3.3.4 Transition temperature. The transition temperature of the silicone sponge rubber (the temperature at which the material changes from a rubber to a rigid solid) shall not be above -56°F (-43°C) for soft grade material or above -60°F (-52°C) for medium grade material. Testing shall be performed in accordance with 4.5.6.

3.3.5 Reactivity. The reactivity between the silicone sponge rubber and AHH propellant shall not exceed 1.0 milliliter (ml). Testing shall be performed in accordance with 4.5.7.

3.4 First article. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.5 Workmanship. Forward cushions shall be fabricated in a thoroughly workmanlike manner and shall be free of large voids or any other obvious material defects.

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Finished parts shall be clean and free of dirt, grease, contamination or any other foreign matter. Cleaning agents shall not be injurious to the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.4), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The sample shall consist of the following items in the sample quantities as indicated.

Item	Drawing and Part No.	Quantity
Specimen of cured silicone sponge rubber (Soft or Medium Grade as applicable) for material analysis	9328352 or 9328353	1 square foot
Forward Cushions (applicable part number)	9299055-1 or 9299055-2	6

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4.3.2 Inspections to be performed. The first article samples submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwgs. and to the inspections and tests specified in the first article inspection table herein. All parts shall have been manufactured using the same production process as will be used in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production and approved in accordance with 3.2. If the first article sample fails to meet any of the applicable requirements, the first article sample shall be rejected.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63169(AR)

PARAGRAPH 4.3.2.1	TITLE Cushion, Forward		SHEET 1 OF 1		DRAWING NUMBER See note b below
					NEXT HIGHER ASSEMBLY M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Material properties	(a)			4.4.2.1
	Classification of Defects and Test (Forward Cushion)	6			4.4.2.1

NOTES:
(a) One square foot specimen of applicable cured silicone sponge rubber.
(b) See dwg. 9299055 and 9328352 or 9328353, as applicable.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots of finished cushions shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot shall consist of items fabricated from a single inspection lot of material produced by a single manufacturer in one continuous process employing not more than one lot of base material and curing agent and approved in accordance with 3.2.

4.4.2 Examinations and tests. See MIL-A-48078 and the quality conformance inspection tables herein. If any sample or specimen fails to meet any of the applicable requirements, the inspection lot which it represents shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63169(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Material Properties of Silicone Sponge Rubber	1 OF 1	See note b below
			NEXT HIGHER ASSEMBLY: 9299055

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Physical properties	(a)		3.3.1	4.5.1 thru 4.5.3
102.	Chemical composition	(a)		3.3.2	4.5.4
103.	Infrared spectrum	(a)		3.3.3	4.5.5
104.	Transition temperature	(a)		3.3.4	4.5.6
105.	Reactivity	(a)		3.3.5	4.5.7

NOTES: (a) One square foot specimen of each applicable cured silicone rubber shall be submitted to the Government (see 6.3) for material property tests and approval prior to use in fabrication of cushions.

(b) See dwg. 9328152 or 9328353, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63169(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cushion, Forward	1 OF 1	9299055
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Outside diameter	4.4. 3.1		3.1	Gage/TMC
102.	Inside diameter of hole	4.4. 3.1		3.1	Gage/TMC
103.	Location of inside diameter of hole	4.4. 3.1		3.1	Gage/TMC
104.	Thickness	(a)		3.1	Gage
Minor:					
201.	Poor workmanship		100%	3.5	Visual

NOTES:
(a) Gage each sheet of material when sheet material is used or one out of each ten cushions if other fabrication methods are used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63169(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Packaging	1 OF 1	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		-	5.0	Visual
102.	Items not clean and free of contamination and foreign material prior to packaging		100%	5.1.1	Visual
103.	Fiberboard separators missing		100%	5.1.1	Visual
104.	Spiral container improper or improperly sealed		100%	5.1.1	Visual
105.	Filler and cushioning does not provide firm pack		100%	5.1.1	Visual/Manual
106.	Exterior pack improper or not properly sealed		100%	5.1.2	Visual
107.	Marking on spiral wound or exterior containers missing, improper or illegible		100%	5.1.3	Visual
108.	Evidence of poor workmanship		100%	-----	Visual

NOTES:

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Tool manufacturing control. When tool manufacturing control (TMC) is specified, the first five pieces initially produced from each mold or die inspected for the applicable classification of defects herein shall constitute initial tool acceptance. Subsequently, the first and last five pieces from each days production inspected for the applicable classification of defects herein shall constitute inspection lot acceptance. In addition, any time the tool requires repair an additional five pieces from each tool shall be inspected for the applicable classification of defects herein before resumption of production. If any sample fails to meet any of the applicable inspection requirements, the lot of production under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" columns in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.4 herein).

4.5 Test methods and procedures.

4.5.1 Density (see 3.3.1). The density of the silicone sponge rubber shall be determined in accordance with ASTM D-792, Method I or an approved equivalent.

4.5.2 Hardness (see 3.3.1). The Shore "A" Durometer hardness of the silicone sponge rubber shall be measured in accordance with ASTM D-2240 or an approved equivalent.

4.5.3 Compression deflection and set (see 3.3.1). The compression deflection and set (recovery) of the silicone sponge rubber shall be determined in accordance with 4.6.1.2 and 4.6.1.3 of MIL-R-6130 or an approved equivalent.

4.5.4 Chemical composition (see 3.3.2). The organic content of the silicone sponge rubber shall be determined at 650°C (the temperature at which most of the decomposable matter is lost) using a DuPont 950 Thermogravimetric Analyzer or an approved equivalent to measure and record the sample weight loss as a function of temperature as shown in Figures 1 and 2 (as applicable). The temperature change shall be 10°C/min. and the medium shall be air.

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4.5.5 Infrared spectrum (see 3.3.3). The infrared spectrum of the silicone sponge rubber shall be determined by pyrolyzing a sample in a test tube over a bunsen burner, transferring the resultant liquid condensate into an infrared transmitting window and recording the infrared spectrum with an infrared spectrophotometer as shown in Figures 3 and 4 (as applicable).

4.5.6 Transition temperature (see 3.3.4). The transition temperature of the silicone sponge rubber shall be determined using a Dupont 941 Thermomechanical Analyzer or an approved equivalent to continuously measure and record the expansion of the material as a function of temperature as shown in Figures 5 and 6 (as applicable). The temperature change shall be 5°C/min. and the medium shall be air.

4.5.7 Reactivity (see 3.3.5). Reactivity tests shall be performed in accordance with MIL-STD-286, Method 408.1.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be semi-bulk packaged in moistureproof spirally wound containers in accordance with MIL-STD-1169, Level A. Items shall be separated by fiberboard separators and fillers and cushioning provided as required.

5.1.2 Packing. Spiral wound containers shall be packed in accordance with MIL-STD-1169, Level B.

5.1.3 Marking. Spiral wound containers and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no
c. Used on M753 Projectile
d. Manufacturer
e. Quantity enclosed
f. Lot no.
g. Contract or purchase order no.
h. Date of acceptance

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6. NOTES

6.1 Intended use. The forward cushions covered by this specification are intended for use in assembly of rocket motors for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type of cushion (see 1.2).
c. First article inspection, when required (see 3.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Test samples. Material samples shall be submitted for approval to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-LCA-OP, Dover, New Jersey 07801 or an approved laboratory.

6.4 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.5 Supersession data. This specification includes the requirements of Purchase Description AR-PD-16 dated 9 April 1979.

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6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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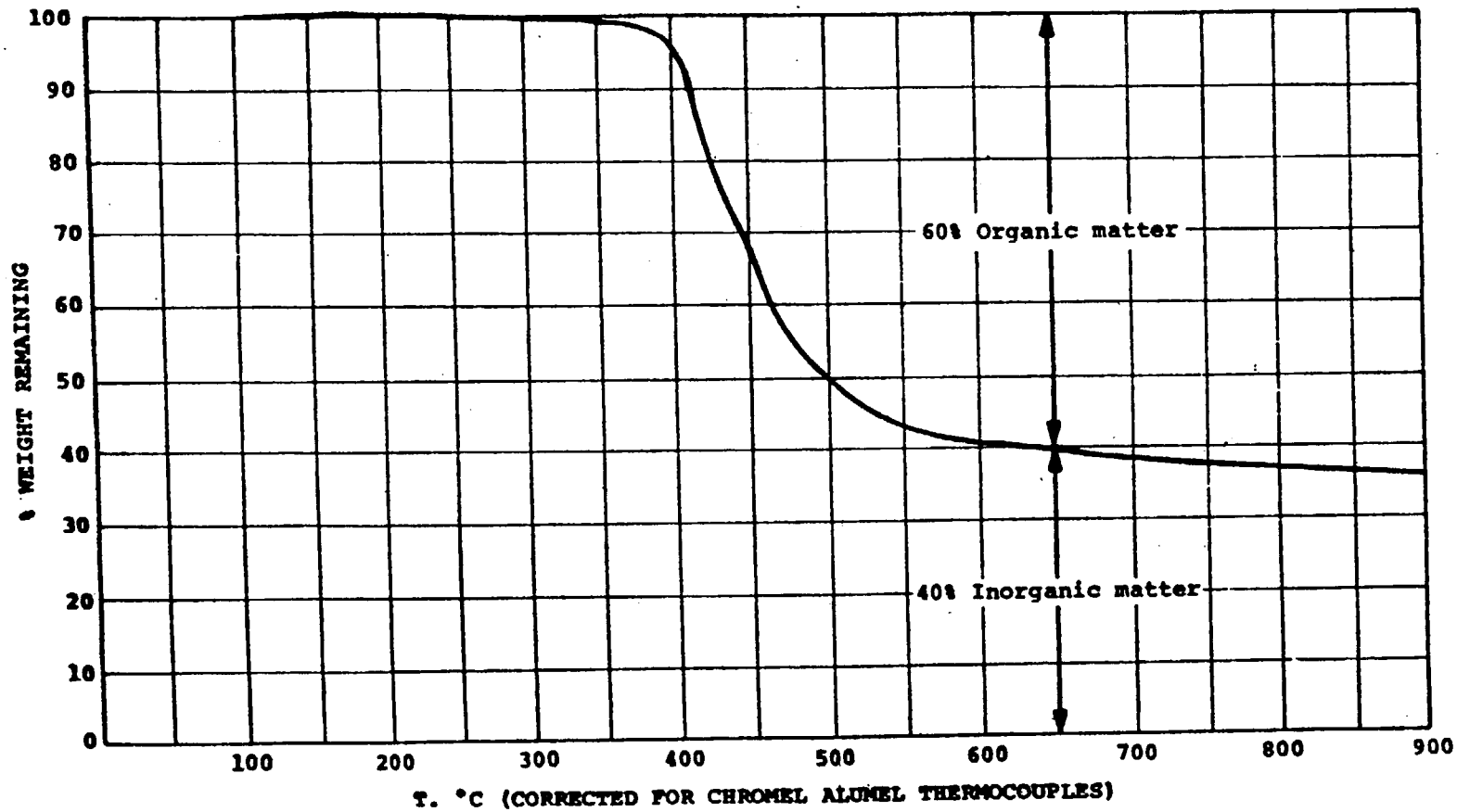


FIGURE 1. Chemical Composition (Soft Grade).

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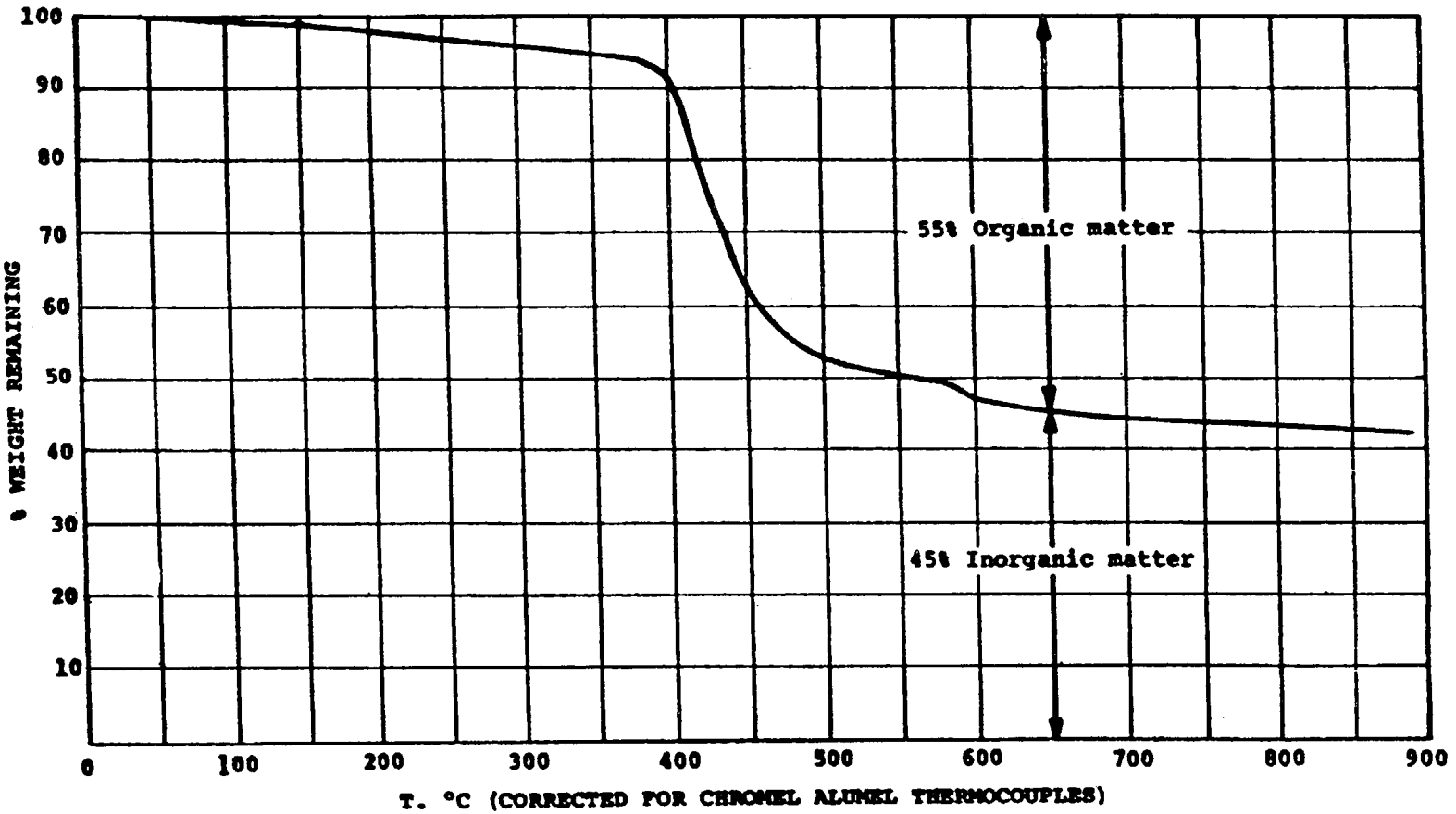


FIGURE 2. Chemical composition (Medium Grade).

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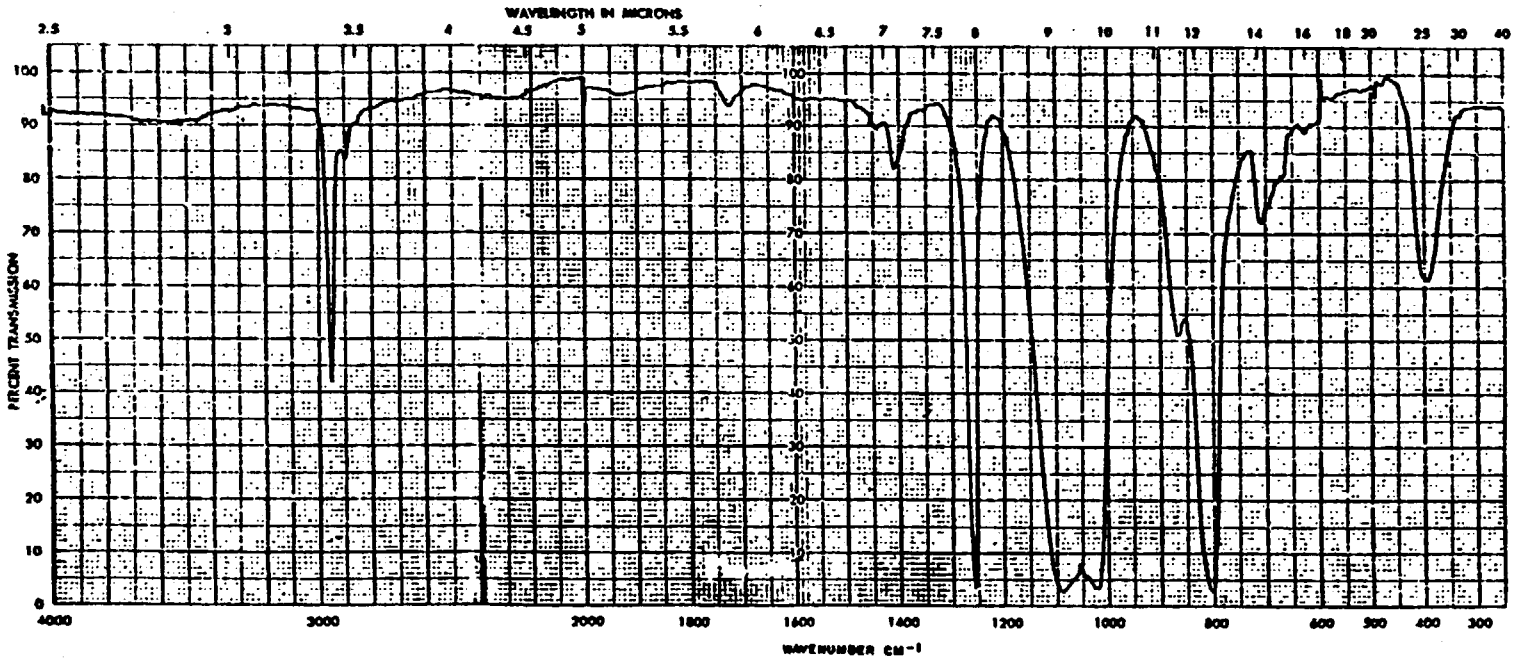


FIGURE 3. Spectrum of Silicone Sponge Rubber (Soft Grade).

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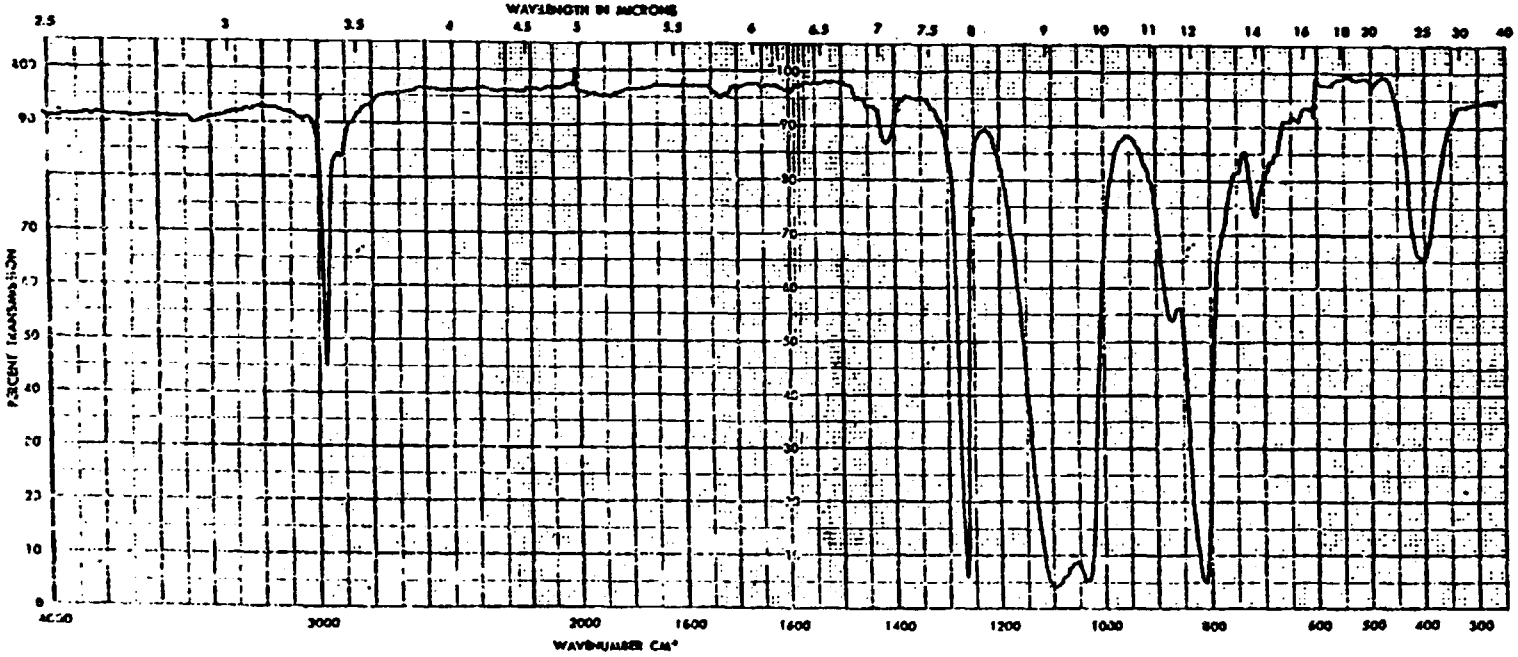


FIGURE 4. Spectrum of Silicone Sponge Rubber (Medium Grade).

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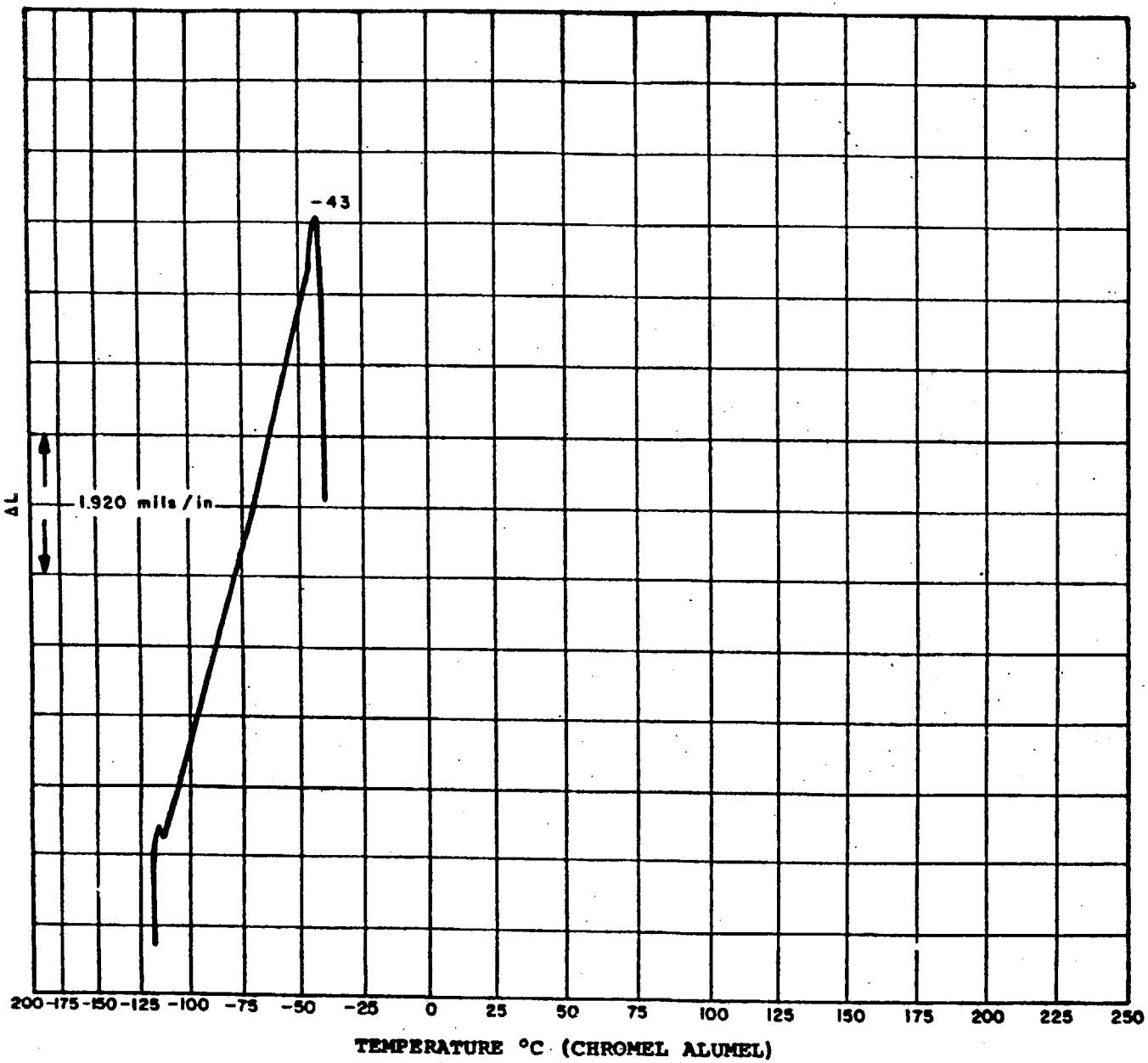


FIGURE 5. Silicone Sponge Rubber Transition Temperature by Thermomechanical Analysis (Soft Grade).

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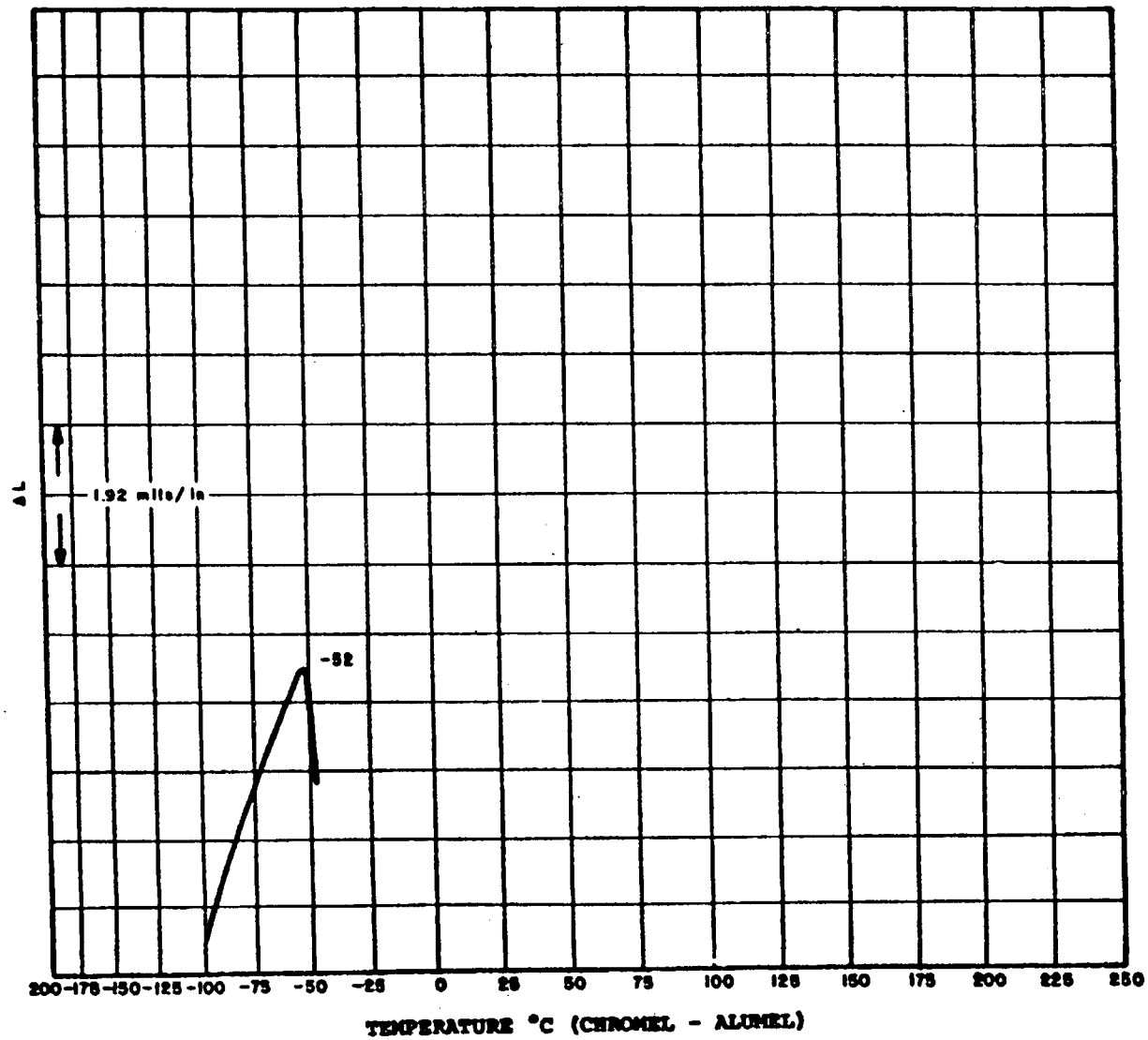


FIGURE 6. Silicone Sponge Rubber Transition Temperature by Thermomechanical Analysis (Medium Grade).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63168 CUSHION, FORWARD M753 8 INCH PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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